

FARIHA SULTANA RINI

Mobile: +8801812320837 | Email: farihasultanarini@gmail.com

Address: Jigatola, Dhaka, Bangladesh

Github: github.com/FarihaSultanaRini

LinkedIn: [linkedin.com/in/fariha-sultana-rini-76b90719a](https://www.linkedin.com/in/fariha-sultana-rini-76b90719a)

PROFESSIONAL SUMMARY

2 years of experience developing ML/AI solutions that turn data into practical business applications. Skilled in end-to-end pipeline from data preprocessing to training models and in turning insights into real-world solutions.

PROFESSIONAL EXPERIENCE

Junior Data Scientist | Somikoron AI

July 2023 - February 2025

Natural Language Processing:

- Data Processing: Built end-to-end Bangla data pipeline for language model and improved data quality by 15% through custom preprocessing functions.
- Traditional NLP Models: Implemented N-Gram and Word2Vec models from scratch and trained them for downstream NLP tasks.
- LLM: Handled tokenization for transformer-based language models for Bangla data, achieving a 5% improvement in vocab quality. Fine-tuned an LLM on our Bangla corpus and extracted and processed embeddings for NLP product development.
- Recommendation Engine: Developed an AI powered Bangla Book recommendation engine.

Computer Vision:

- Data Processing: Annotated and preprocessed data for object detection and instance segmentation.
- YOLO: Trained YOLOv5 model and achieved 10% accuracy improvement through data augmentation, hyperparameter tuning.

Common Crawl Data Pipeline:

- Built and maintained data pipelines to filter and preprocess Bangla text from large-scale multilingual Common Crawl datasets for AI model development.

Audio Signal Processing:

- Preprocessed audio data and developed a frequency-level sound mixer for AI voice cloning projects.

Data Science Intern | Somikoron AI

December 2022 - June 2023

- Performed web scraping, data preprocessing and exploratory data analysis.
- Built OCR pipelines for text extraction using PyTesseract and EasyOCR.

Research Assistant | Aachol Foundation

July 2022 - October 2022

- Collected, preprocessed, and analyzed mental health data.

EDUCATION

Bachelor of Science in Computer Science and Engineering ,

University of Asia Pacific | CGPA-3.58 / 4.0 | Graduation Year : June 2022

Higher Secondary School Certificate ,

Birshreshtha Noor Mohammad Public College | GPA : 5.0 / 5.0 | Year : 2017

Secondary School Certificate ,

Birshreshtha Noor Mohammad Public College | GPA : 5.0 / 5.0 | Year : 2015

RESEARCH EXPERIENCE AND PUBLICATION

Paper Title: Predicting Optimum Lockdown Pattern of Epidemic Spread Using AI Techniques

Authors: Fariha Sultana Rini, Somiya Assir Jebin, Merajunnesa Kumkum, et al.

Journal: Research Briefs on Information Communication Technology Evolution (ReBICTE), Volume 9, 2023.

SKILLS

Programming Languages: Python(NumPy,Pandas,Seaborn,Matplotlib,Scikit-learn,Pytorch)

NLP: BERT, LLaMA, Word2Vec, N-Gram, FAISS, Hugging Face Transformers, NLTK, RAG

OCR: PyTesseract, EasyOCR, Mistral OCR

Computer Vision: YOLOv5(Object Detection, Instance Segmentation), Roboflow, CVAT

MLOps: Docker, Apache Airflow, MLflow, Amazon Bedrock

API Development: FastAPI, RESTful APIs

Version Control: Git, GitHub

ACADEMIC PROJECTS

Thesis: AI-based Covid-19 Model | **Tech stack:** NumPy, Pandas, Matplotlib, Scikit-learn, MESA

Spring 2021-Fall 2021

Predict optimum lockdown pattern of epidemic spread using AI techniques.

GitHub

Flood Alert System | **Tech Stack:** Arduino IDE, C++, Arduino Uno, GSM, Water Sensor, Buzzer

Fall 2020

A flood alarm system that sends signals according to the water level to avoid sudden flood-related accident. GitHub

Vangariwala | **Tech Stack:** Python, Django, Html, CSS, Bootstrap

Spring 2020

A website that provides a platform where people can sell their unused things from their homes, and offices. GitHub

Bank Management System | **Tech Stack:** Java

Spring 2019

Using this system, users can deposit money, withdraw money, or update their account information.

GitHub

Student Management System | **Tech Stack:** Microsoft SQL Server

Fall 2019

A student management system that allows storing, updating, deleting, or creating of student data.

GitHub

ACADEMIC AWARDS

Dean's List

Fall 2021, Spring 2021

Merit Waiver

Fall 2020, Spring 2020, Fall 2019

EXTRACURRICULAR ACTIVITIES

UAP CSE Hardware and Software (SnH) club

2019

Assisted in event planning for Hardware and Software Carnival

Inter-Department Programming Contest

2018

Provided guidance to participants and assisted with the setup of the software

ENTREPRENEURIAL COMPETITIONS

Unibator

Participated in the Unibator's startup competition. Out of 7152 teams,my team was chosen as one of the top 250.

Black Brains

Taken part in Ideation 1.0, a Visionary Idea Competition run by Black Brains, and qualified till the 2nd round.